VisionGuard Accessibility Manual

Complete Reference: Touch Gestures, Assistive Actions & Spoken Confirmations

Hybrid Edge + Gemini AI

Zero-Vision Touch Screen Architecture: The entire screen functions as an active blind-friendly gesture surface. Users do not need to target buttons or look at the display. Every gesture triggers immediate haptic feedback and clear spoken voice confirmation via speech synthesis.

👆 1. One-Finger Navigation & Controls

GESTURE	ASSISTIVE ACTION	SPOKEN VOICE CONFIRMATION	PEDESTRIAN USE CASE
Single Tap 1 finger tap	**Toggle Scanning** Pauses or resumes live camera obstacle tracking and sonar pings.	`"Scanning active"` `"Scanning paused"`	Pause while talking to someone, riding an elevator, or seated.
Double Tap 2 rapid taps (1 finger)	**Repeat Hazard Report** Re-analyzes and repeats current urgent obstacles or clear path.	`"Obstacle report: [hazards]"` `"Path is clear ahead"`	When approaching a curb, crossing, or doorway to double-check.
Swipe Up 1 finger flick up	**Toggle Flashlight / Torch** Powers the rear camera LED torch for low-light vision.	`"Flashlight on" / "Flashlight off"`	Walking on unlit sidewalks, dim stairwells, or evening navigation.
Swipe Down 1 finger flick down	**Announce System Status** Speaks battery %, inference FPS, active mode & obstacle count.	`"VisionGuard active. Battery 85%, running at 4 frames per second. 1 obstacle detected."`	Periodic battery & system health check without sighted assistance.
Swipe Right 1 finger flick right	**Cycle Feedback Audio Mode** Cycles audio verbosity: Full ➔ Voice Only ➔ Sonar Only ➔ Emergency Only.	`"Full audio mode" / "Voice only mode"` `"Sonar only mode" / "Emergency only mode"`	Quiet transit mode (Emergency only) vs noisy streets (Full audio).
Swipe Left 1 finger flick left	**Toggle High-Contrast Theme** Instantly toggles Obsidian Dark and Porcelain Light themes.	`"Switched to light mode" / "Switched to dark mode"`	Adapting to bright outdoor sunlight glare vs indoor lighting.
Long Press Hold finger down 1.5s	**Cycle Verbosity (Alternative)** Secondary trigger for cycling audio feedback modes.	`"Full audio mode" / "Voice only mode"`	Convenient when wearing winter gloves or holding cane.

✌️ 2. Two-Finger Multimodal AI & Assistive Intelligence

GESTURE	ASSISTIVE ACTION	SPOKEN VOICE CONFIRMATION	PEDESTRIAN USE CASE
Two-Finger Tap 2 fingers tap once	🧠 **Google Gemini Scene Narrator** Snaps high-res frame and speaks natural spatial layout.	`"Analyzing scene with Google AI..."` `"At 11 o'clock is an open door, at 12 o'clock is a desk, at 2 o'clock is an office chair."`	Entering an unfamiliar room, hotel lobby, restaurant, or intersection.
Two-Finger Double Tap 2 fingers double tap	**Re-check Telemetry & Path** Re-announces telemetry and surrounding context.	`"VisionGuard active. Full audio mode. 0 obstacles detected."`	Reassurance that system is functioning during quiet stretches.
Two-Finger Swipe Up 2 fingers flick up	📖 **Instant Text & Sign Reader (OCR)** Continuous OCR scanning text every 3.5s with speech.	`"Text Reader active. Point camera at text."` `"Text reader stopped"`	Finding room numbers, reading street signs, product labels, or notices.
Two-Finger Swipe Down 2 fingers flick down	💡 **Light & Window Tone Detector** Audio oscillator maps light (0–255) to pitch (180–1100 Hz).	`"Light tone active. Pitch rises near light sources."` `"Light tone off"`	Locating daylight windows, confirming lights are on/off before leaving.
Two-Finger Swipe Right 2 fingers flick right	🎯 **Target Finder Next Object** Cycles target forward: door ➔ chair ➔ stairs ➔ elevator ➔ crosswalk.	`"Target set to door. Point camera to locate."` `"Target set to stairs. Point camera to locate."`	Locating building exits, empty seats on a bus/subway, or staircases.
Two-Finger Swipe Left 2 fingers flick left	🎯 **Target Finder Previous Object** Cycles target backward through library.	`"Target set to crosswalk..."`	Quickly returning to a previous target without cycling all options.
Two-Finger Long Press Hold 2 fingers 1.5s	**Open / Close Settings Panel** Opens accessible configuration dialog for speed, volume, and AI.	`"Settings opened"` `"Settings closed"`	Changing voice rate, lens FOV preset, or checking API connectivity.

🚨 3. Three-Finger Safety Gesture

GESTURE	ASSISTIVE ACTION	SPOKEN VOICE CONFIRMATION	PEDESTRIAN USE CASE
Three-Finger Tap 3 fingers tap screen	🛑 **EMERGENCY PAUSE (Kill Switch)** Instantly halts all audio pings, speech synthesis, and engines.	`"VisionGuard emergency pause. Tap screen to resume."`	Emergency vehicle sirens approaching, crossing busy street, hearing safety.

VisionGuard Features & Architecture Guide

Engine Architecture, Audio Radar Bands & Zero-Typing Setup

Hybrid System

4. On-Screen HUD Quick Actions (For Sighted Testers & Low-Vision Users)

Describe Scene Button

Snaps active camera frame and invokes Google Gemini 2.0 Flash to synthesize a detailed spatial orientation paragraph with clock directions.

Read Text (OCR) Button

Activates automatic sign and text reading. Point phone at signs, food packages, or mail. Deduplicates repeated text automatically.

Light Tone Button

Starts continuous audio pitch oscillator. High frequency whistle indicates facing a bright light source or daylight window.

Seek Target Button

Cycles search targets (door, chair, stairs, etc.). Emits 3D binaural directional beacon sound that centers when facing target.

Cash / Currency Button

Snaps frame and identifies banknote currency and denomination (e.g. "*Twenty US Dollars*") to prevent payment errors.

Settings & Torch Buttons

Top HUD bar gives direct access to flashlight toggle, dark/light theme switch, pause/resume, and settings configuration dialog.

5. 3D Spatial Audio Radar & Proximity Band Guide

PROXIMITY BAND	ESTIMATED DISTANCE	SPATIAL SONAR AUDIO CUE	VOICE SPEECH BEHAVIOR
CLEAR / SAFE	Beyond 3.5 meters	Low-frequency intermittent pulse (1.2s cadence)	Ambient path status spoken every 15 seconds.
CAUTION	1.5 to 3.5 meters	Medium-pitch pulse (500ms cadence), pans stereo left/right	Speaks obstacle class and zone: "*Chair right caution*".
DANGER / CRITICAL	Under 1.2 meters (cane reach)	High-pitch rapid beacon (150ms cadence) + haptic buzz	**Immediate audio ducking & interrupt:** "*Stop! Person center danger*".

6. Zero-Typing Project Configuration via .env

Instant Google Gemini AI Configuration

You never need to type or paste an API key on your mobile phone screen. Configure it directly in the project file:

- Open `.env` in the project root on your computer.
- Set: `VITE_GEMINI_API_KEY=your_gemini_api_key_here`
- Get a free key (no credit card required) at **aistudio.google.com**.
- When deploying to **Vercel**, add `VITE_GEMINI_API_KEY` under Project Settings ➔ Environment Variables. Every mobile phone visiting your Vercel URL will automatically load the key!

7. Testing Checklist for New Users

1 Test Edge Collision

Walk toward a chair or wall. Hear the sonar tempo speed up and pan left/right. At 1.2m, verify the critical interrupt: "*Stop! [Object] danger*".

2 Test Gemini Scene Narrator

Double-tap with two fingers in any room. Listen to Gemini describe furniture, doors, and clear paths using clock directions.

3 Test Text Reader

Swipe up with two fingers. Aim camera at a book, food box, or street sign. Hear the speech synthesizer read the words.

4 Test Light Detector

Swipe down with two fingers. Aim camera toward a window or lamp. Notice the oscillator whistle rise into a high pitch.

Safety Notice: VisionGuard is an electronic orientation aid and does NOT replace a physical white cane or guide dog.